

Solax Microinverter Communication



Solax Microinverter Communication Solution

Solax Microinverter applies the WiFi wireless communication and Power line communication for different microinverter series. For Solax X1-Micro 2 in 1 and X1-Micro 4 in 1 series, they can not directly communicate with ECC(Energy Control Center) as shown in the following figure 1, though ECC and X1-Micro 2 in 1 and X1-Micro 4 in 1 series microinverter have built-in WiFi modules. In this scenario, ECC is not necessary, but if the microinverter system needs to achieve zero export or interact with external devices (such as heat pumps etc.), ECC is required. Both of microinverter and ECC will connect to the router and upload their data to the Solax Cloud on the hour.

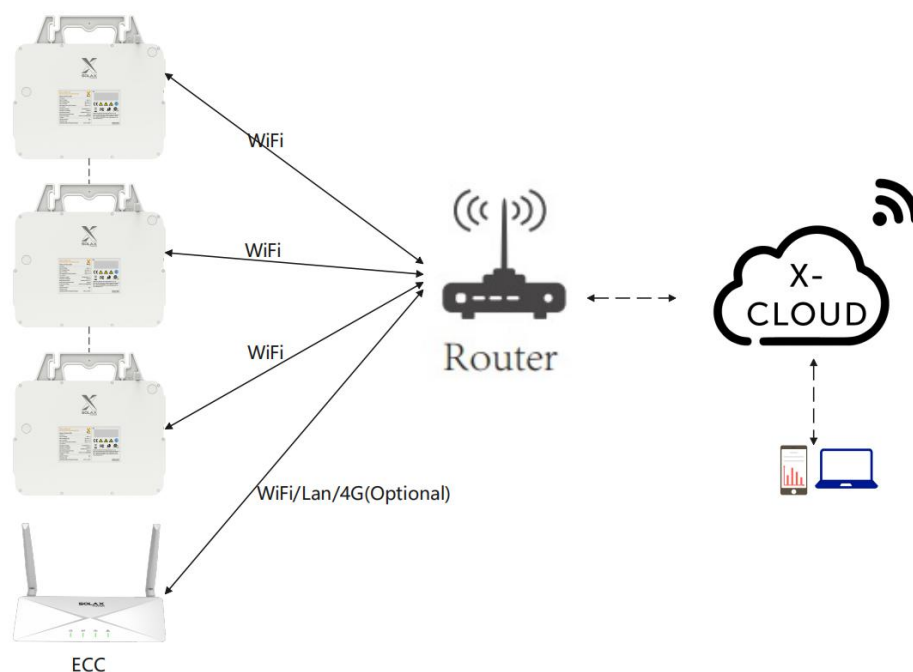


Figure1

For Solax A1-Micro 1 in 1 series, ECC-PLC supports bidirectional communication which can collect the microinverter power generation data and then send these data to ECC that will transfer it to SolaxCloud, and send remote control commands to the microinverter via power line communication (put the communication signal over the AC electrical conductors).

Both ECC-PLC and A1-Micro 1 in 1 series microinverter have industrial-grade built-in PLC module. Also, microinverter can not only directly communicate with the ECC-PLC as shown in figure 2, but also enable seamless remote maintenance and system adjustments.

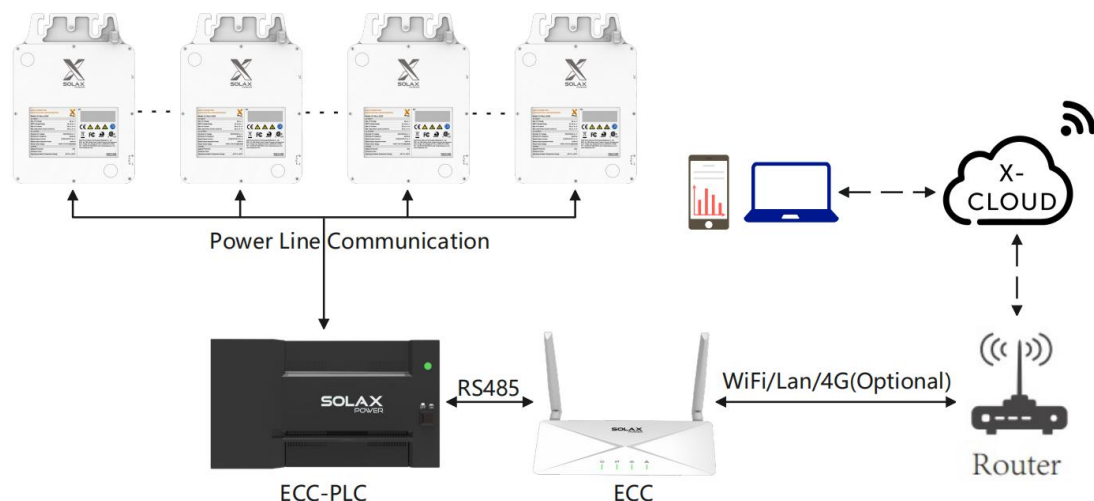


Figure2

	PLC	Sub-1G	WiFi	Zigbee
Communication Speed	Low	Low	High	Low
Communication Node	Many	Many	Few	Many
Communication security	User-defined	User-defined	High	High
Energy Consumption	Low	Average	High	Low
Data transfer rate	2Mbps	50kbps	100Mbps	250Kbps
Features	Communication data carry in the same cable, saving the cost of expensive outdoor cables; Noise problem affecting the quality of communication	Applicable to the scenario with many obstacles; Support mesh network; Low data transfer rate;	Can directly access the Internet, no gateway needed; relatively fast data transfer rate;	Applicable to large- scale data acquisition and control; Support mesh network;

Comparison of typical communication method

Application of WiFi/PLC Communication in Microinverter System

A microinverter system has two options for communicating with the monitoring system: either through an independent communication gateway or through integrated Wi-Fi module that directly accesses the internet. If you are constructing a compact residential solar power system that requires only a handful of microinverters, opting for microinverters equipped with integrated Wi-Fi represents the most economical choice. Using integrated Wi-Fi Microinverter also simplifies the installation of smaller solar systems. While independent gateways must be installed indoors after the rest of the installation is completed, microinverters with integrated Wi-Fi eliminate this additional component; everything is installed on the rooftop, obviating the need for a gateway inside the building.

However, when installing a large-scale system that utilizes numerous microinverters, an independent gateway is likely the most efficient solution. With its expansive monitoring range, an independent gateway can seamlessly handle data transfer from a substantial number of microinverters without becoming overwhelmed.